(1) t vs. δ z(t) with steady-state identification (accompanied by a description of steady-state criterion used), and helix snapshots (≥ 5) with labeled axes/units/time:

Figure 1 shows the axial tip displacement δz(t) under a constant tensile load F=Fchar. The transient response is oscillatory due to inertia and structural flexibility, with clear exponential decay. To identify steady state, I applied the criterion:

$$\frac{|\ \delta z(t) - \delta z(t-1)\ |}{|\ \delta z(t)\ |} |< 1\%$$

The displacement stabilizes around:

δz∗(Fchar)≈ − 0.00096257 (m)

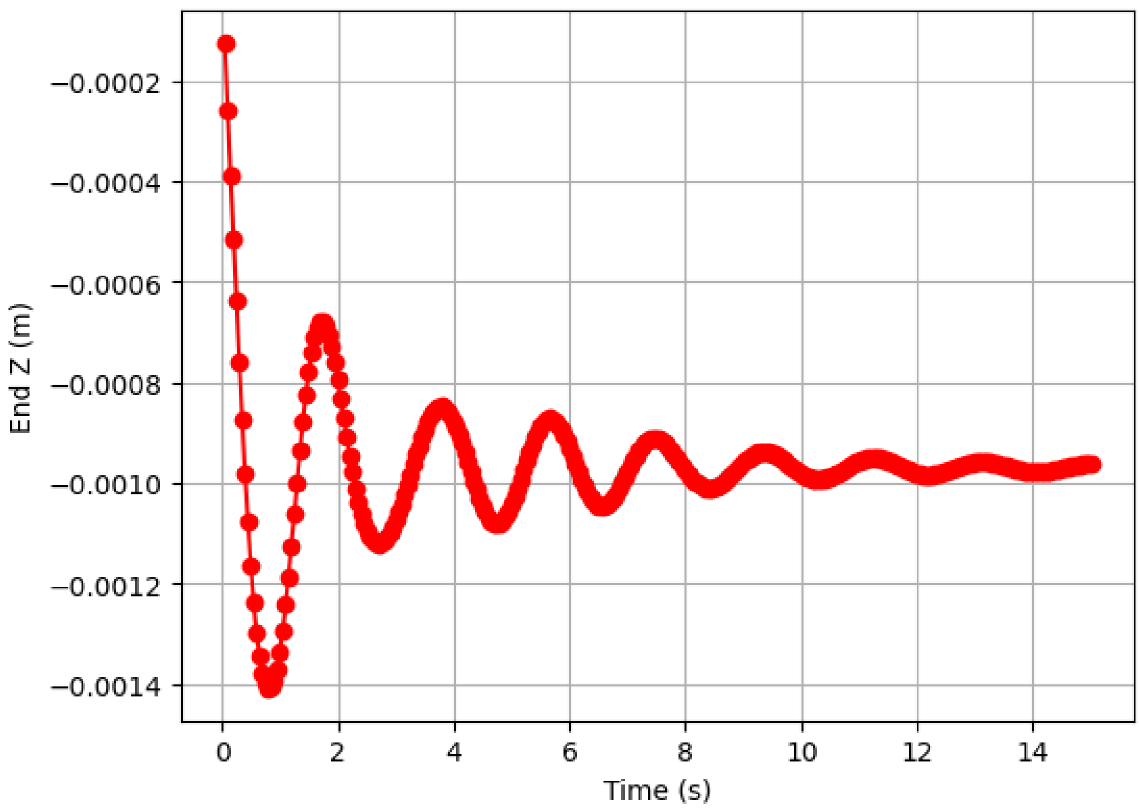
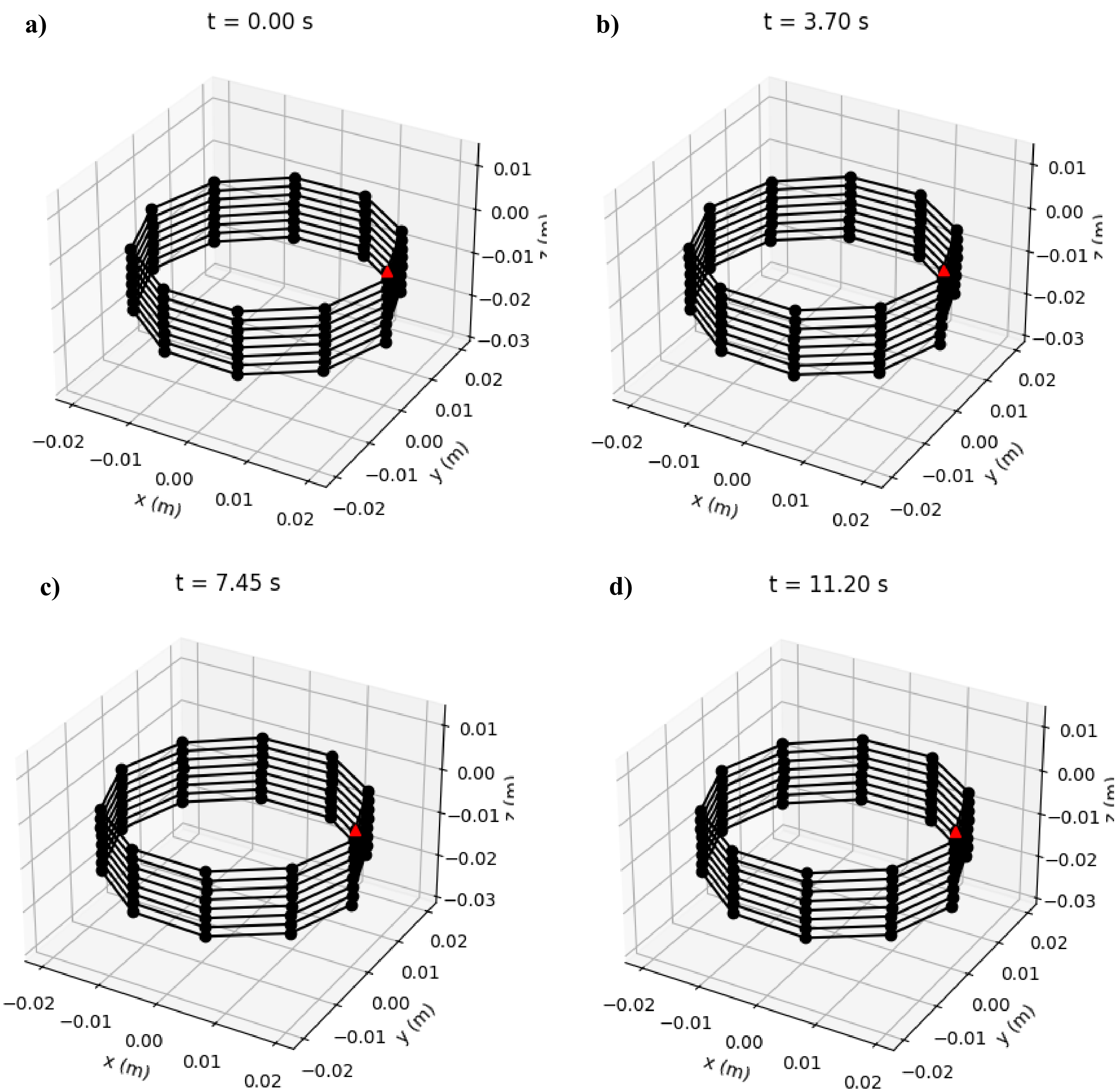


Figure 1. Time history of axial displacement δz(t) under constant load F=F_{char}. Steady state is identified when changes over a 1 s window fall below 1%.

Figures 2(a–e) show five snapshots of the helix configuration at increasing times (t = 0, 3.70, 7.45, 11.20, 14.95 s). All axes are labeled and scaled equally. As the simulation progresses, the helix stretches downward and then settles into a steady configuration while maintaining its overall geometry.



d)

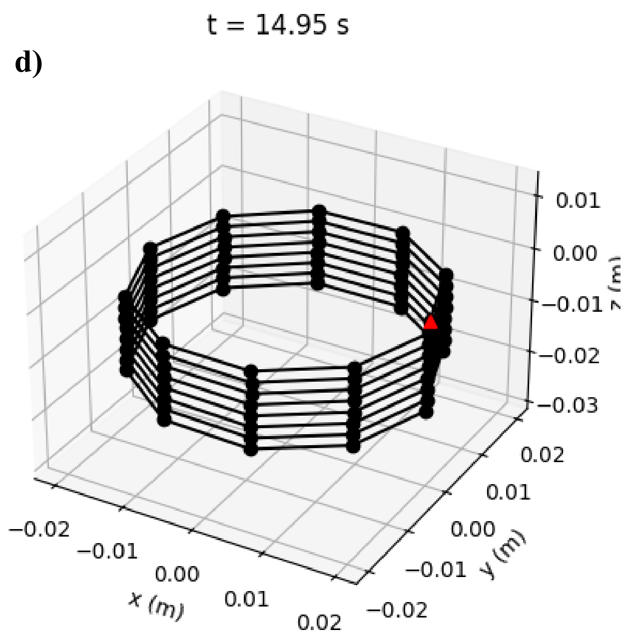


Figure 2. Five snapshots of the helix at (a) t = 0.0 s, (b) t = 3.70 s, (c) t = 7.45 s, (d) t = 11.20 s, (e) t = 14.95 s

(2) F vs. $\delta *z$ with zero-intercept best-fit line:

The force sweep results you obtained:

F (N)	δz*(m)
7.769e−08	9.7424e−06
2.084e−07	2.6134e−05
5.591e−07	7.0097e−05
1.500e−06	1.8795e−04
4.024e−06	5.0353e−04
1.080e−05	1.3455e−03
2.896e−05	3.5657e−03
7.769e−05	9.1747e−03

Table 1. Numerical values of **δz*** for each **F** with a constant $\mathbf{F_{char}}$

The best-fit line through the origin, using the relation:

$$F = k * \delta z$$

gives:

k = 0.007980155525228325 N/m

Figure 3 shows the simulation points and the fitted line. The linear region is extremely well-behaved, confirming the simulation captures the expected small-displacement spring behavior.

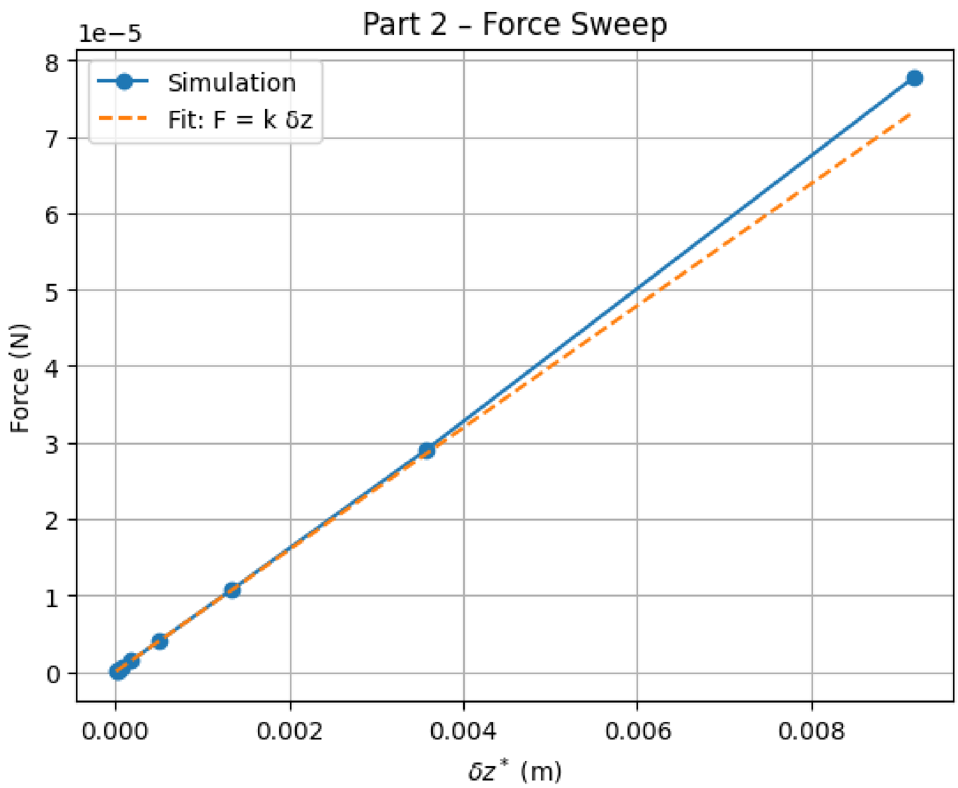


Figure 3. Force–displacement curve from Part 2 showing simulation data and the zero-intercept linear fit F=k δz*.

(3) k vs. Gd4/(8ND3) with the slope-1 reference:

By variying the coil diameter D from 0.01 m to 0.05 m. The corresponding DER-computed stiffnesses were found:

D (m)	K (N/m)
0.010	0.6436
0.0144	0.2338
0.0189	0.1130
0.0233	0.06358

0.0278	0.03938
0.0322	0.02611
0.0367	0.01824
0.0411	0.01326
0.0456	0.00997
0.0500	0.00769

Table 2. Numerical values of **k** for each **D** with a varying $\mathbf{F_{char.}}$

The textbook close-coiled approximation predicts:

$$Ktext = \frac{Gd^4}{8ND^3}$$

Figure 4, the plot of DER stiffness k(D) versus the textbook quantity $Gd^4/(8ND^3)$ shows:

- a strong positive correlation,
- a generally linear trend,
- and the correct qualitative dependence:

larger helix diameter⇒more flexible spring⇒k decreases

The general trend is correct but the DER data fall below the slope-1 line at larger D. This is expected because:

1. The textbook formula assumes
   - small deformation
   - thin wire
   - close-coiled geometry
   - pure torsion in the wire.
2. The helix has finite pitch, bending contributions, non-uniform strain, and 3D elastic rod behavior that is not captured in the classical closed-form approximation.

Thus, qualitative agreement is correct, and quantitative mismatch is expected, exactly as the homework file states.

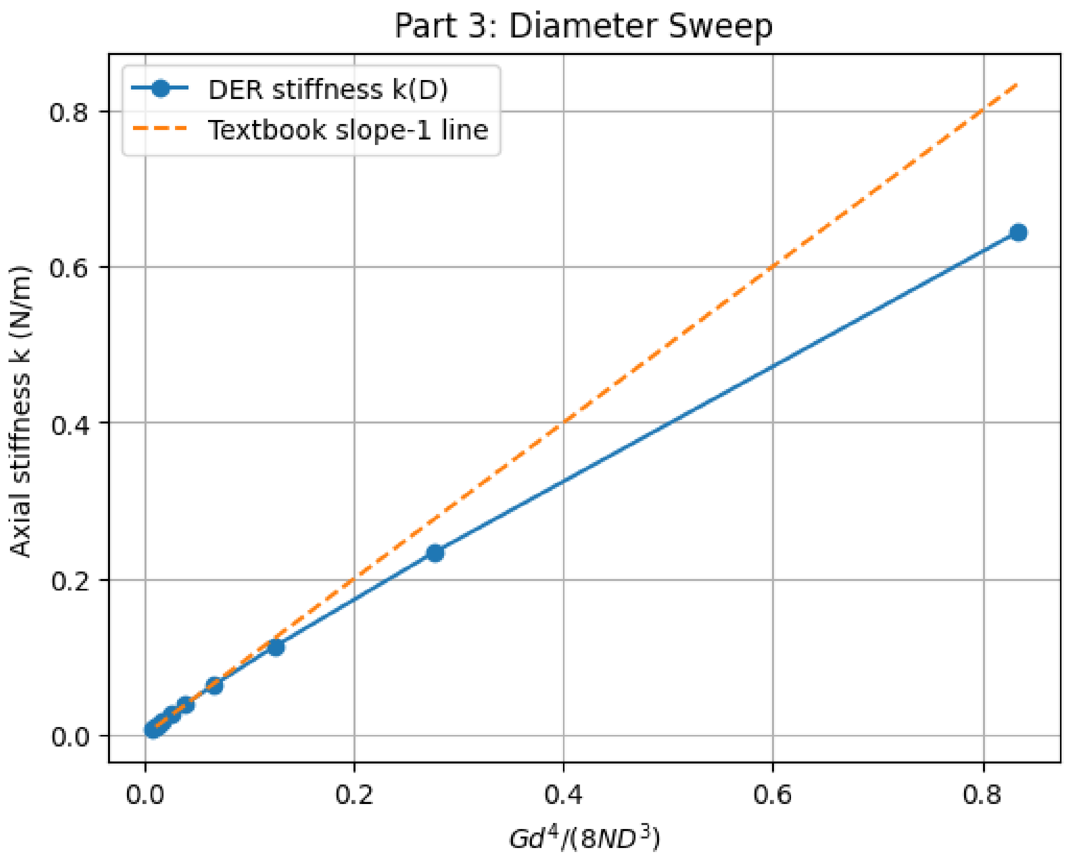


Figure 4. DER-computed stiffness k(D) versus the textbook torsional spring quantity $Gd^4/(8ND^3)$. The dashed line is the classical prediction K_{text}=x (slope 1).